

# One-operation completeness for polynomial rings: a characteristic-two dichotomy

— condensed version —

John Thompson

June 2026

## Abstract

Call  $p \in R[x, y]$  *complete* if every bivariate polynomial over  $R$  is an iterated composition of  $p$ , the variables, and constants. We prove: a complete binary polynomial exists over a commutative ring  $R$  if and only if 2 is a unit of  $R$ ; when it is,  $x^2 - y$  is complete, and when it is not, for every  $p$  at least one of  $x + y$ ,  $xy$  is inexpressible.

## 1 Definitions and statements

**Definition 1.** Let  $R$  be a commutative ring and  $p \in R[x, y]$ . The *substitution clone*  $\text{Clo}(p) \subseteq R[x, y]$  is the smallest set containing  $x$ ,  $y$ , and every constant  $c \in R$ , and closed under  $(f, g) \mapsto p(f, g)$  (substitution of  $f$  for  $x$  and  $g$  for  $y$  in  $p$ ). Call  $p$  *complete* over  $R$  if  $\text{Clo}(p) = R[x, y]$ .

Two points of precision. First,  $\text{Clo}(p)$  is not a clone of operations on a set: its elements are *formal* polynomials — it is the binary component of the subclone of the abstract clone (Lawvere theory) of commutative  $R$ -algebras generated by  $p$ , the projections, and the constants; equivalently, the subalgebra of the magma  $(R[x, y], (f, g) \mapsto p(f, g))$  generated by  $\{x, y\} \cup R$ . Second, the formal/functional distinction has teeth:  $n := 1 + xy \in \mathbb{F}_2[x, y]$  induces NAND on  $\{0, 1\}$  — a single-operation analogue of the Sheffer stroke [2] — so the *function* clone it generates contains every Boolean function, yet  $Dn = 1 \neq 0$  for the operator  $D$  of Section 3, so Theorem 4 gives  $x + y \notin \text{Clo}(n)$ : no composition of  $n$  equals the polynomial  $x + y$  on the nose.

**Theorem 2** (dichotomy). *Let  $R$  be a commutative ring. A complete binary polynomial over  $R$  exists if and only if 2 is a unit of  $R$ .*

**Theorem 3.** *If  $\frac{1}{2} \in R$  then  $q(x, y) := x^2 - y$  is complete:  $\text{Clo}(x^2 - y) = R[x, y]$ .*

**Theorem 4.** *For every  $q \in \mathbb{F}_2[x, y]$ ,  $x + y \notin \text{Clo}(q)$  or  $xy \notin \text{Clo}(q)$ .*

There is no hypothesis on  $q$ : reduction mod 2 collapses degree  $(2x^3 + xy \mapsto xy)$ , so the deduction of Corollary 6 below requires Theorem 4 for arbitrary  $q$ , degenerate cases included.

**Theorem 5.** *For every field  $k$  of characteristic 2 and every  $q \in k[x, y]$ , at least one of  $x + y$ ,  $xy$  lies outside  $\text{Clo}(q)$ .*

**Corollary 6.** *No binary polynomial over  $\mathbb{Z}$  is complete.*

**Lemma 7.**  $\text{Clo}(p) = R[x, y]$  if and only if  $x + y \in \text{Clo}(p)$  and  $xy \in \text{Clo}(p)$ .

*Proof.* Substituting members of the clone into a member lands in the clone (induction on the formula). So if  $x + y, xy$  are members, the clone is closed under  $+$  and  $\times$ ; containing the variables and constants, it is everything.  $\square$

**Lemma 8** (transfer). *A ring homomorphism  $R \rightarrow S$  induces  $\pi : R[x, y] \rightarrow S[x, y]$  with  $\pi(\text{Clo}(p)) \subseteq \text{Clo}(\pi p)$ .*

*Proof.*  $\pi$  fixes variables, sends constants to constants, and commutes with substitution; induct over the formula.  $\square$

By Lemmas 7 and 8 applied to  $\mathbb{Z} \rightarrow \mathbb{F}_2$ , Theorem 4 implies Corollary 6.

We use three facts about the formal partials  $\partial_x, \partial_y$  on  $\mathbb{F}_2[x, y]$ : (1)  $\partial(f^2) = 2f \partial f = 0$ ; (2)  $\partial_x^2 = \partial_y^2 = 0$ , while  $D := \partial_x \partial_y$  need not vanish; (3) squaring is a ring homomorphism, and a polynomial with all exponents even is a square. Univariate degrees follow the convention that constants — including 0 — have degree 0, so  $\deg(f^2) = 2 \deg f$  without exception.

## 2 The positive half: $x^2 - y$

*Proof of Theorem 3.* By Lemma 7 it suffices to reach  $x + y$  and  $xy$ . Everything flows from

$$q(\alpha, q(\alpha', \gamma)) = \alpha^2 - \alpha'^2 + \gamma. \quad (1)$$

*Translations.* With constants  $\alpha = \frac{d+1}{2}$ ,  $\alpha' = \frac{d-1}{2}$ , (1) sends  $\gamma \mapsto \gamma + d$  for any  $d \in R$  (the only step using  $\frac{1}{2}$ ); in particular  $x + d, y + d \in \text{Clo}(q)$ .

*Slanted lines.*  $q(c, y) = c^2 - y$ , then  $q(x, c^2 - y) = x^2 + y - c^2$ , then

$$q(x + d, x^2 + y - c^2) = 2dx - y + (d^2 + c^2) \in \text{Clo}(q),$$

a line of slope  $2d$  for every  $d$ , constant adjustable by translations.

*Linear engine.* For  $\alpha \in \text{Clo}(q)$  and constant  $k$ , (1) with outer argument  $\alpha + k$  and inner argument  $\alpha$  gives the increment

$$\gamma \mapsto \gamma + 2k\alpha + k^2.$$

Increments along slanted lines of two distinct slopes span all linear polynomials, so from  $\gamma = 0$ : every linear polynomial lies in  $\text{Clo}(q)$ , in particular  $x + y$  and  $x + \frac{y}{2}$ .

*Scaled squares.* The engine with  $\alpha = y^2 + \ell$  (in the clone via  $q(y, -\ell)$ ) adds  $\lambda y^2 + \text{const}$  for any  $\lambda$ ; from  $\gamma = x^2 = q(x, 0)$  we get  $x^2 + \frac{1}{4}y^2 \in \text{Clo}(q)$ .

*Finish.*

$$q\left(x + \frac{y}{2}, x^2 + \frac{1}{4}y^2\right) = \left(x + \frac{y}{2}\right)^2 - x^2 - \frac{1}{4}y^2 = xy. \quad \square$$

### 3 The negative half over $\mathbb{F}_2$

Fix  $q \in \mathbb{F}_2[x, y]$ . Grouping the monomials of  $q$  by the parities of their exponents and using fact (3),  $q$  has a unique *parity decomposition*

$$q = A_0^2 + xA_1^2 + yA_2^2 + xyA_3^2, \quad A_i \in \mathbb{F}_2[x, y], \quad (2)$$

and differentiating, with squares derivative-free,

$$\partial_x q = A_1^2 + yA_3^2, \quad \partial_y q = A_2^2 + xA_3^2, \quad Dq = A_3^2. \quad (3)$$

**Lemma 9** (chain rule for  $D$ ). *In characteristic 2, for all  $q, \alpha, \beta \in \mathbb{F}_2[x, y]$ ,*

$$D(q(\alpha, \beta)) = (Dq)(\alpha, \beta) (\partial_x \alpha \partial_y \beta + \partial_y \alpha \partial_x \beta) + (\partial_x q)(\alpha, \beta) D\alpha + (\partial_y q)(\alpha, \beta) D\beta.$$

*Proof.* Expand  $\partial_x \partial_y (q(\alpha, \beta))$  by the chain and product rules; the terms involving  $\partial_x^2 q$  or  $\partial_y^2 q$  vanish by fact (2), and the survivors assemble as displayed.  $\square$

**Proposition 10.** *If  $Dq = 0$  then  $xy \notin \text{Clo}(q)$ .*

*Proof.*  $\{g : Dg = 0\}$  contains the variables and constants and, by Lemma 9 with the first term dead, is closed under substitution into  $q$ ; hence it contains  $\text{Clo}(q)$ . But  $D(xy) = 1$ .  $\square$

Suppose henceforth  $Dq \neq 0$ , i.e.  $A_3 \neq 0$ ; we show  $x + y \notin \text{Clo}(q)$ . Write  $\mathbb{F}_2[x+y]$  for the subring of polynomials in  $x + y$ . The proof runs through the rigidity property

( $\dagger$ ) for all  $\alpha, \beta \in \mathbb{F}_2[x, y]$ : whenever  $q(\alpha, \beta)$  is a nonconstant element of  $\mathbb{F}_2[x+y]$ , both  $\alpha, \beta \in \mathbb{F}_2[x+y]$ .

**Proposition 11** (master reduction). *If  $q$  satisfies ( $\dagger$ ) then  $x + y \notin \text{Clo}(q)$ .*

*Proof.* We claim every  $g \in \text{Clo}(q)$  satisfies  $P(g) : g \notin \mathbb{F}_2[x+y]$  or  $g$  constant. The substitution  $x \leftrightarrow y$  fixes  $\mathbb{F}_2[x+y]$  pointwise but moves  $x$  and  $y$ , so the variables satisfy  $P$ ; constants do. Let  $g = q(\alpha, \beta)$  with  $P(\alpha), P(\beta)$  and suppose  $g \in \mathbb{F}_2[x+y]$  is nonconstant. Then ( $\dagger$ ) forces  $\alpha, \beta \in \mathbb{F}_2[x+y]$ ;  $P$  forces them constant; then  $g$  is constant — contradiction. So  $P$  holds on  $\text{Clo}(q)$ , and  $x + y$  fails  $P$ .  $\square$

Fix an algebraically closed field  $K \supseteq \mathbb{F}_2$  of characteristic 2 containing an element  $\sigma$  transcendental over  $\mathbb{F}_2$ ; concretely  $K = \overline{\mathbb{F}_2(\sigma)}$ . Call  $c \in K$  *transcendental* if it satisfies no nonzero polynomial relation over  $\mathbb{F}_2$ .

**Definition 12.** A *witness* is a pair  $\alpha, \beta \in K[t]$ , not both constant, with  $q(\alpha, \beta) = c$  constant and  $c$  transcendental.

**Proposition 13** (bridge). *If no witness exists, then  $q$  satisfies ( $\dagger$ ).*

*Proof.* Let  $q(\alpha, \beta) = f(x + y)$  with  $f \in \mathbb{F}_2[u]$  nonconstant. Substitute  $(x, y) = (t, t + \sigma)$ : the right side becomes the constant  $f(\sigma)$ , which is transcendental (a nonzero  $\mathbb{F}_2$ -relation on  $f(\sigma)$  composes with  $f$  to one on  $\sigma$ ). Since no witness exists,  $\alpha(t, t + \sigma)$  and  $\beta(t, t + \sigma)$  are constant in  $t$ . Finally, a  $g \in \mathbb{F}_2[x, y]$  constant along the generic line lies in  $\mathbb{F}_2[x+y]$ : expanding  $g(x, x + u) = \sum_i g_i(u) x^i$ , constancy gives  $g_i(\sigma) = 0$  for  $i \geq 1$ , hence  $g_i = 0$ , so  $g = g_0(x + y)$ .  $\square$

It remains to prove:

**Theorem 14** (no witnesses). *If  $Dq \neq 0$ , no witness exists.*

## 4 The Frobenius descent

Throughout,  $'$  denotes  $d/dt$  and degrees are degrees in  $t$ . Over  $K[t]$ : squares have zero derivative, and conversely a polynomial with zero derivative is a square (it is a polynomial in  $t^2$ , and the coefficients have square roots since  $K$  is perfect).

Set  $w := \gcd(A_1, A_2, A_3)$  and  $A_i = wB_i$ , so the  $B_i$  have no common nonunit factor and

$$q = A_0^2 + w^2 h, \quad h := xB_1^2 + yB_2^2 + xyB_3^2, \quad (4)$$

with  $B_3 \neq 0$  and partials  $h_x = B_1^2 + yB_3^2$ ,  $h_y = B_2^2 + xB_3^2$ ,  $Dh = B_3^2$ . The peel (4) removes the possible curve of critical points of  $q$  along  $\{w = 0\}$ ; Lemma 17 shows none survives in  $h$ . Three lemmas, then the descent.

**Lemma 15** (kill lemma). *Let  $(\alpha, \beta)$  be a witness with level  $c$ , and  $f \in \mathbb{F}_2[x, y]$ ,  $f \neq 0$ . Then  $f(\alpha, \beta) \neq 0$  in  $K[t]$ .*

*Proof sketch.* Suppose  $f(\alpha, \beta) = 0$ ; after swapping variables assume  $\alpha$  nonconstant. View  $f$  and  $g := q + z$  as polynomials in  $y$  over  $\mathbb{F}_2[x, z]$ : since  $g$  is monic and linear in  $z$  it is irreducible and cannot divide the  $z$ -free  $f$ , so they are coprime, and a Bézout identity with denominators cleared gives  $fU + (q + z)V = E(x, z) \neq 0$  in  $\mathbb{F}_2[x, z][y]$ . Substituting  $y \mapsto \beta$ ,  $x \mapsto \alpha$ ,  $z \mapsto c$  kills both summands ( $q(\alpha, \beta) + c = 0$ ), so  $E(\alpha, c) = 0$ ; expanding  $E = \sum_i \rho_i(z) x^i$  and using that  $1, \alpha, \alpha^2, \dots$  have strictly increasing degrees in  $t$ , every  $\rho_i(c) = 0$ , and some  $\rho_i \neq 0$  exhibits  $c$  as algebraic over  $\mathbb{F}_2$  — contradiction.  $\square$

**Lemma 16** (algebraic points). *If  $u, v \in \mathbb{F}_2[x, y]$  have no common nonunit factor and  $(a, b) \in K^2$  satisfies  $u(a, b) = v(a, b) = 0$ , then  $a, b$  are algebraic over  $\mathbb{F}_2$ .*

*Proof sketch.* In  $\mathbb{F}_2(x)[y]$  a common divisor would descend to  $\mathbb{F}_2[x, y]$  by Gauss's lemma [1, Ch. IV], so  $u, v$  are coprime there; a cleared Bézout identity gives  $su + tv = r(x) \neq 0$ , and evaluation at  $(a, b)$  gives  $r(a) = 0$ . Symmetrically for  $b$ .  $\square$

**Lemma 17** (keystone). *Let  $B_1, B_2, B_3 \in \mathbb{F}_2[x, y]$  have no common nonunit factor, with  $B_3 \neq 0$ , and set  $h_x := B_1^2 + yB_3^2$ ,  $h_y := B_2^2 + xB_3^2$ . Then  $h_x$  and  $h_y$  have no common nonunit factor.*

*Proof.* Both are nonzero: applying  $\partial_y$  to a relation  $h_x = 0$  would give  $B_3^2 = 0$ . Suppose a prime  $p$  divides both.

If  $p \mid B_3$ : subtracting the  $B_3^2$ -terms,  $p$  divides  $B_1^2$  and  $B_2^2$ , hence (primality) all three  $B_i$  — excluded.

If  $p \nmid B_3$ : write  $h_x = pu$  and  $h_y = pv$ . Differentiate, remembering squares are derivative-free:

$$0 = \partial_x h_x = p_x u + p u_x \implies p \mid p_x u;$$

$$B_3^2 = \partial_y h_x = p_y u + p u_y \implies p \nmid u$$

(else  $p \mid B_3^2$ , so  $p \mid B_3$ );

$$B_3^2 = \partial_x h_y = p_x v + p v_x \implies p \nmid p_x$$

(same reason). So the prime  $p$  divides  $p_x u$  but neither factor — contradiction.  $\square$

*Proof of Theorem 14.* Among all witnesses take one minimizing  $n := \deg \alpha + \deg \beta$  ( $\geq 1$ , as not both are constant); fix  $r := \sqrt{c}$ , transcendental (if  $r$  were algebraic, so were  $c = r^2$ ). By Lemma 15 applied to the nonzero polynomials  $w, h_x, h_y, B_3$ , none of

$$W := w(\alpha, \beta), \quad A := h_x(\alpha, \beta), \quad B := h_y(\alpha, \beta), \quad \Delta := B_3(\alpha, \beta)^2$$

vanishes in  $K[t]$ .

*Peel.* From (4) and  $q(\alpha, \beta) = c = r^2$ , using  $(u + v)^2 = u^2 + v^2$ ,

$$W^2 H = S^2, \quad H := h(\alpha, \beta), \quad S := r + A_0(\alpha, \beta); \quad (5)$$

differentiating ( $W^2$  and  $S^2$  are squares) gives  $W^2 H' = 0$ , so  $H' = 0$ .

*Engine.* The chain rule for  $h$ , with  $\partial_x h_x = \partial_y h_y = 0$  and  $\partial_y h_x = \partial_x h_y = Dh = B_3^2$ , gives

$$A\alpha' + B\beta' = H' = 0, \quad A' = \Delta\beta', \quad B' = \Delta\alpha',$$

whence  $(AB)' = \Delta(B\beta' + A\alpha') = 0$  and  $AB$  is a square in  $K[t]$ .

*Fork, branch 1:*  $A$  and  $B$  share a root  $\theta \in K$ . Then  $P := (\alpha(\theta), \beta(\theta))$  is a common zero of  $h_x, h_y$ , which by Lemmas 17 and 16 has algebraic coordinates. Evaluating (5) at  $\theta$ ,  $S(\theta)^2 = w(P)^2 h(P)$  is algebraic, hence so is  $S(\theta)$ , hence so is  $r = S(\theta) + A_0(P)$  — contradicting transcendence of  $r$ .

*Fork, branch 2:*  $A, B$  coprime. Coprime factors of a square are squares (units of  $K[t]$  are constants of the algebraically closed  $K$ , hence squares), so  $A' = B' = 0$ ; since  $\Delta \neq 0$ , the engine identities force  $\alpha' = \beta' = 0$ , so  $\alpha = \alpha_1^2, \beta = \beta_1^2$  with  $\deg \alpha_1 + \deg \beta_1 = n/2$ . Because the coefficients of  $q$  lie in  $\mathbb{F}_2$ , squaring commutes with substitution into  $q$ :

$$q(\alpha_1, \beta_1)^2 = q(\alpha_1^2, \beta_1^2) = c,$$

so  $q(\alpha_1, \beta_1) = r$  and  $(\alpha_1, \beta_1)$  is a witness of total degree  $n/2 < n$  — contradicting minimality. (If  $n = 1$  halving is impossible and this branch cannot occur, so the descent is well-founded.)  $\square$

Theorem 14 gives  $(\dagger)$  by Proposition 13, hence  $x + y \notin \text{Clo}(q)$  by Proposition 11 when  $Dq \neq 0$ ; with Proposition 10 this proves Theorem 4, and with Lemma 8, Corollary 6.

## 5 From $\mathbb{F}_2$ to the dichotomy

**Proposition 18.** *For every perfect field  $k$  of characteristic 2 and every  $q \in k[x, y]$ , at least one of  $x + y, xy$  lies outside  $\text{Clo}(q)$ .*

*Proof.* The descent runs over  $k$  with one change: coefficients of  $q$  are no longer fixed by squaring, so in the halving branch  $q(\alpha_1^2, \beta_1^2) = q^{(1/2)}(\alpha_1, \beta_1)^2$ , where  $q^{(1/2)}$  applies the inverse Frobenius to each coefficient (perfectness enters here); the halved pair is a witness for  $q^{(1/2)}$ , and since the induction is on witness degree with  $q$  universally quantified inside, the twist is absorbed. Parity decomposition, bridge, and master reduction are unaffected.  $\square$

*Proof of Theorem 5.*  $k$  embeds into its perfection  $k^{1/2^\infty}$ ; by Lemma 8, membership of both  $x + y$  and  $xy$  in  $\text{Clo}(q)$  over  $k$  would persist over the perfection, contradicting Proposition 18.  $\square$

*Proof of Theorem 2.* If 2 is a unit,  $x^2 - y$  is complete (Theorem 3). If not, 2 lies in a maximal ideal  $\mathfrak{m}$ ; a complete operation would push forward along  $R \rightarrow R/\mathfrak{m}$  (Lemma 8) to one over the residue field, of characteristic 2 — contradicting Theorem 5.  $\square$

All results have been machine-verified in Lean 4 against Mathlib (axioms: propositional extensionality, quotient soundness, choice); the development accompanies the paper.

Three questions remain open. Which degrees  $\geq 3$  admit complete operations over  $\mathbb{Q}$  ( $x^3 - y$  is a candidate; over  $\mathbb{F}_3$  its clone consists of additive-plus-constant polynomials, so it is not complete there)? Which  $p$  are complete when 2 is a unit ( $xy + 1$  and  $x^2 + y$  are not, by a characteristic-zero instance of  $(\dagger)$  and by a positivity obstruction, respectively)? Is membership in  $\text{Clo}(p)$  — or completeness — decidable? The identity  $q(f, q(f, g)) = g$  for  $q = x^2 + y$  over  $\mathbb{F}_2$  shows substitution admits unbounded cancellation, so bounded search proves nothing.

## References

- [1] S. Lang, *Algebra*, rev. 3rd ed., Graduate Texts in Mathematics 211, Springer, New York, 2002.
- [2] H. M. Sheffer, *A set of five independent postulates for Boolean algebras, with application to logical constants*, Trans. Amer. Math. Soc. 14 (1913), 481–488.